

6. MASS AND BALANCE

6.1 EQUIPMENT LIST ENTRIES

System/Equipment Designation	Part No.	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Pulsed Chip Detector System	105-98267	1.5	3600	5400

7. SYSTEM DESCRIPTION

A **MAG PLUG** warning light coming on is an indication of conditions which could cause engine failure. The pulsed chip detector system is designed to eliminate annoying **MAG PLUG** warning lights caused by normal wear conditions.

Normal engine component wear (due to gear mesh, bearing rotation, etc.) creates fine soft metal particles that mix with oil or soft carbon to form a paste that collects on the magnetic chip detectors. Activation of the pulsed chip detector system sends a stored electrical pulse to each chip detector to burn off this paste. However, large metal particles that indicate component failure are not burned off by the system as shown by the **MAG PLUG** warning light remaining on.

The pulsed chip detector system comprises four pulsed chip detectors (two per engine), a power module, and a guarded spring-loaded PULSED CHIP DET switch.

When enough metal particles collect on an engine's magnetic detectors to close the circuit, the respective **MAG PLUG** warning light will come on.

The Main Bus, PP21 supplies 28V DC power to the power module via the PULSED CHIP DET circuit breaker on the center console switch panel. Placing the PULSED CHIP DET switch on the overhead panel to the forward, on position triggers the power module to release its capacitor-stored charge that arcs across each of the four chip detectors. If the **MAG PLUG** warning light goes off, the collected particles have been eliminated.